

ASSESSING KNOWLEDGE, ADOPTION WILLINGNESS, AND SKILL ACQUISITION PREFERENCES FOR GENERATIVE AI AMONG EMPLOYEES IN SMALL TO MEDIUM-SIZED ENTERPRISES

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1 INTRODUCTION

As artificial intelligence (AI), particularly generative AI, continues to transform industries, businesses are seeking ways to integrate AI into their operations to enhance productivity and remain competitive. For small to medium-sized enterprises (SMEs), the challenge is not only adopting generative AI, but also ensuring that their workforce is adequately equipped to leverage these technologies effectively.

Employees in these organizations may have varying degrees of familiarity with generative AI, from little to no exposure. The roles of these employees could benefit from a basic understanding of using generative AI tools, such as text generation, image generation and other technologies. Given the constraints such as limited resources, time, and access to formal training, it is essential for SMEs to identify effective methods for teaching AI to employees.

This study aims to identify effective methods for teaching generative AI to employees. By understanding the employees' current knowledge levels, willingness to learn, and preferred learning methods, SMEs can create effective strategies to teach their employees and capitalize on the potential of generative AI.

1.1 Research Question

What is the current level of knowledge about generative AI among employees in small to medium-sized companies, how willing are they to adopt it in their roles, and what is their preferred way of acquiring these skills?

To provide an answer to this research question, we have come up with the following sub-questions:

1. How do negative concerns about Artificial Intelligence relate to SME employees' willingness to learn about Artificial Intelligence.
2. What is the relationship between the willingness among employees in SMEs to adopt generative AI in their roles and their age?
3. What factors contribute to employees' concerns and misconceptions about AI in the workplace?
4. How is an SME employee's existing knowledge of generative AI associated with their willingness to pursue further learning about the usage of generative AI?

1.2 Problem Statement

The adoption of generative AI in SMEs may be influenced by a lack of understanding about the current level of knowledge and willingness of employees to adopt this technology in their roles as well as their preferred methods for acquiring the necessary skills. This knowledge gap affects not only the employees themselves, who are concerned about job security

and the relevance of their skills, but also SME owners and managers, who are responsible for making strategic decisions about AI adoption and are concerned about return on investment, employee retention, and competitive advantage.

Furthermore, the successful integration of generative AI in SMEs also depends on the support of industry associations and chambers of commerce, government and policymakers, educational institutions and training providers, AI and technology developers, and customers and clients of SMEs. These stakeholders have varying levels of power, attention, and interest in generative AI adoption, but all play a critical role in shaping the ecosystem that enables or hampers the adoption of generative AI in SMEs.

By investigating the current level of knowledge, willingness, and preferred learning methods among employees in SMEs, this research aims to provide valuable insights that can inform strategies for successful adoption and integration of generative AI in these organizations, ultimately benefiting all stakeholders involved.

2 LITERATURE STUDY

Research done in one of our found sources shows that when adopting AI into SMEs, things like employee attitudes, organizational culture, and the needed resources all influence the willingness of employees within small to medium sized companies to learn about and integrate AI technologies into their domains. The paper clarifies that SMEs face many challenges such as limited financial and personnel resources, that can ease or hinder AI training efforts (Schwaeke and Jones (2024)). Their systematic-literature review of 106 peer-reviewed articles on AI adoption in SMEs, provides valuable insights and clarifies AI implementation within SMEs.

The paper focuses on the point that knowledge and training are important factors in the success of AI integration in the SME space. Employees' positive attitudes and digital proficiency significantly affect the smoothness of AI adoption. When it benefits and aligns with the SME and employee needs and innovation objectives, the likelihood of their enthusiastic adoption increases. Furthermore, a strategic investment in human resources, including AI education and upskilling, is vital for increasing employee willingness to embrace AI. So, looking at how SME employees can be educated on this is important. Overall, these insights point to the direction for a need of AI training that both addresses technical skills and employee concerns about job security and the future of their roles in the world of AI.

In a new research paper we chose, the author explores factors influencing the adoption of generative AI in businesses using the Technology- Organization-Environment framework. Six major factors are identified as having a significant impact on adoption: compatibility, complexity, organizational size, regulatory support, environmental uncertainty, and competition intensity. The paper acknowledges the fact that integrating Generative AI into businesses can provide a big advantage, like enhancing decision- making and innovation which could become crucial for the competitive advantage. But doing so presents numerous hurdles, which is why organizations are sometimes hesitant to fully integrate it, they are wary of the complexities and uncertainties surrounding this technology, but larger organizations with more resources and less hurdles to face are more likely to adopt these technologies. The study highlights that complexity, such as the lack of standards, significantly hinders adoption, while compatibility with existing systems facilitates smoother integration, as it needs to be compatible with existing processes to facilitate the AI integration. Overall, the paper emphasizes that strategic investments in technology infrastructure, human resources, and regulatory alignment are necessary to overcome barriers and successfully adopt generative AI (Agrawal (2023)).

3 METHODOLOGY

3.1 Research Method and Approach

This study will utilize both qualitative and quantitative research methods to assess how employees in SMEs can be effectively taught about AI development. A mixed-methods approach allows for a better understanding of employees' attitudes, knowledge, and preferences regarding AI development in their roles.

- Qualitative Research: In-depth interviews will be conducted with 5 to 8 SME employees to gather nuanced insights

into their understanding of AI, the challenges they face, and their learning preferences. This will provide context for the quantitative data and highlight individual experiences or concerns.

- **Quantitative Research:** A structured survey, distributed via Qualtrics, will gather data on a larger scale. This survey will target a broad audience of SME employees globally, with the possibility of expanding to include larger enterprises or non-SME participants depending on coordination with other research teams. The survey will include questions designed to assess four key areas:
 1. **Current level of AI knowledge:** Measuring participants' self-assessed knowledge of AI concepts and tools.
 2. **Willingness to learn:** Evaluating how motivated employees are to engage with AI learning opportunities.
 3. **Interest in specific AI topics:** Identifying which areas of AI (e.g., data analysis, machine learning, ethics) employees are most interested in learning about.
 4. **Preferred learning methods:** Understanding employees' preferred formats for acquiring new skills, such as in-person workshops, online courses, or self-paced learning.

3.2 Data Collection Methods

Primary data will be collected using the following methods:

- **Surveys:** The primary tool will be a survey distributed through Qualtrics. Through this survey, quantitative (multiple-choice and scaled responses) data will be collected. The survey will be sent out in September 2024.
- **Interviews:** Our research teams' known people and other, through the process found, SME employees will be invited to participate in follow-up interviews, allowing for deeper exploration of their perspectives on AI development and its relevance to their work.
- **Secondary Data:** Existing literature and research papers on AI training, adoption, and employee development in SMEs will be reviewed to support and contextualize the findings from the primary data.

3.3 Sampling Procedures

Basic random sampling will be used to select participants for the survey. The Qualtrics platform will randomly distribute the survey to SME employees around the world, ensuring a diverse and representative sample of participants. This approach will help generalize the results across SMEs, providing insights into employee perspectives on AI learning and adoption. By relying on random sampling, we aim to capture potential differences in employee tendencies and preferences based on factors such as company size and interest in AI learning, without introducing bias in participant selection.

3.4 Data Management Plan

The data management plan will outline how survey responses and interview transcripts will be securely stored, processed, and analysed, adhering to data privacy regulations such as GDPR.

3.5 Data Analysis Techniques

- **Descriptive Statistics:** Summary statistics will be used to analyse quantitative survey data, providing insights into overall trends, such as employee interest in generative AI, preferred learning methods, and perceived barriers to AI adoption.
- **Data Visualization:** Graphs, charts, and tables will be used to visually represent findings from the survey, making it easier to compare preferences and attitudes across different employee groups.

- Inference: Inference techniques like hypothesis testing will help determine whether the differences observed in employee willingness or AI knowledge across different groups are statistically significant and not due to random chance.

3.6 Limitations

Potential limitations of the study include the self-reported nature of survey data, which may introduce biases such as social desirability bias. Additionally, given the global distribution of the survey, cultural and regional differences in AI adoption may affect responses. These factors will be considered when interpreting the results.

4 PREDICTED OUTCOMES

Four possible outcomes have been predicted for this study, guided by hypotheses that aim to explore employees' interest in AI technologies, learning preferences, barriers to AI adoption, and the perceived impact of AI on job performance.

The first hypothesis predicts that employees who are more comfortable with AI tools will show greater interest in learning AI development. Based on this, we expect that most employees will demonstrate interest in learning generative AI, especially those who already recognize its relevance to their roles. AI is becoming more integrated into business processes, and employees likely understand the importance of keeping up with these developments.

The second hypothesis posits that employees who identify time constraints as a barrier will prefer self-paced learning options over in-person workshops. As a result, we predict that many employees will favor flexible, self-guided learning formats such as online courses, with fewer opting for in-person workshops. Those unfamiliar with AI tools may feel slightly uncomfortable, which could further motivate them to choose learning methods that allow for flexibility.

The third hypothesis suggests that employees who believe AI will improve their job performance will be more willing to dedicate 3-5 hours or more per week to learning about AI. We expect that time constraints and the complexity of AI will be the primary barriers to learning, with fewer employees citing a lack of interest or management support. Given the workload of employees in small and medium-sized businesses, finding time for additional learning may be challenging.

Finally, the fourth hypothesis states that employees with less confidence in applying AI concepts will see the complexity of AI as a significant barrier to learning. We predict that many employees will believe learning AI skills will enhance their job performance, especially those with higher confidence in using AI. This belief in AI's ability to improve productivity and career prospects will likely drive a strong interest in acquiring more AI-related skills.

The variables being assessed in this survey include key demographic and AI-related factors. Age is measured on an ordinal scale, representing discrete age groups to examine generational trends. Country of residence is captured as a nominal variable, identifying the respondent's location. Willingness to adopt AI is an ordinal variable, rated from "not interested" to "very interested." Current AI knowledge is self-assessed on an ordinal scale from 1 to 100, showing familiarity with specific AI tools. Preferred learning method and AI subjects of interest are nominal variables that reflect respondents' preferred learning formats and AI topics of interest. Lastly, job role captures respondents' professional roles as a nominal variable, helping to analyze AI adoption across different job functions.

The findings from this survey will provide valuable insights for designing generative AI training programs tailored to employees in small to medium-sized companies. By addressing common barriers such as time constraints and complexity, and offering learning methods that align with employees' preferences, companies can equip their workforce with essential AI skills, ultimately improving productivity and innovation.

Hypotheses:

1. Employees who are more comfortable with AI tools will show greater interest in learning AI development.
2. Those who identify time constraints as a barrier will prefer self-paced learning options over in-person workshops.

3. Employees who believe that AI will improve their job performance will be more willing to dedicate 3-5 hours or more per week to learning about AI.
 4. Employees with less confidence in applying AI concepts will see the complexity of AI as a significant barrier to learning.
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1. How do negative concerns about Artificial Intelligence relate to SME employees' willingness to learn about Artificial Intelligence
 - H_0 : Negative concerns about Artificial Intelligence have no significant influence on SME employees' willingness to learn about Artificial Intelligence.
 - H_A : Negative concerns about Artificial Intelligence have a significant positive influence on SME employees' willingness to learn about Artificial Intelligence.
 2. What is the relationship between the willingness among employees in SMEs to adopt generative AI in their roles and their age?
 - H_0 : There is no relationship between the age of employees in small to medium-sized companies and their willingness to adopt generative AI in their roles.
 - H_A : There is a relationship between the age of employees in small to medium-sized companies and their willingness to adopt generative AI in their roles.
 3. What factors contribute to employees' concerns and misconceptions about AI in the workplace?
 - H_0 : The explored factors have no influence on the concerns and misconceptions employees have about AI.
 - H_A : The explored factors contribute to the concerns and misconceptions employees have about AI.
 4. How is an SME employee's existing knowledge of generative AI associated with their willingness to pursue further learning about the usage of generative AI?
 - H_0 : There is no relationship between existing generative AI knowledge of a SME employee and their willingness to pursue further learning about the usage of generative AI.
 - H_A : There is a relationship between existing generative AI knowledge of a SME employee and their willingness to pursue further learning about the usage of generative AI.

REFERENCES

Agrawal, K. P. (2023). Towards adoption of generative ai in organizational settings. *Journal of Computer Information Systems*.

Schwaeke, J., P. A. K. D. K. K. S. and Jones, P. (2024). The new normal: The status quo of ai adoption in smes. *Journal of Small Business Management*, pages 1–35.

APPENDIX 1: SURVEY QUESTIONS

Section 1: Demographics

1. How do you describe yourself?
 - ☐ Male
 - ☐ Female
 - ☐ Non-binary / third gender
 - ☐ Prefer to self-describe:
 - ☐ Prefer not to say
2. How old are you?
 - ☐ 10-19
 - ☐ 20-29
 - ☐ 30-39
 - ☐ 40-49
 - ☐ 50-59
 - ☐ 60-69
 - ☐ 70-79
 - ☐ Other:
3. What is your country of residence?
4. How many employees does your company have?
 - ☐ 0-10 employees
 - ☐ 11-50 employees
 - ☐ 50-250 employees

Section 2: Current Level of Generative AI knowledge

1. There are sufficient opportunities within my organization to learn about AI
 - ☐ Strongly disagree
 - ☐ Disagree
 - ☐ Neutral
 - ☐ Agree
 - ☐ Strongly agree
2. How often do you use AI tools or technologies in your work?
 - ☐ Not at all
 - ☐ Less than once a month
 - ☐ Once a month
 - ☐ Several times a month
 - ☐ Once a week
 - ☐ Several times a week
 - ☐ Every day

- ☐ Multiple times per day
3. How familiar are you with the following AI tools on a scale from 1 to 100?
- Chatbots (ChatGPT, Gemini)
- Image generation (DALL-E, Midjourney)
- Speech generation (Descript, ElevenLabs, Resemble.ai)
4. How comfortable are you with the idea of integrating AI tools into your daily work tasks?
- ☐ Very uncomfortable
- ☐ Somewhat uncomfortable
- ☐ Neutral
- ☐ Somewhat comfortable
- ☐ Very comfortable
5. How confident are you in your ability to use generative AI in your current job responsibilities?
- ☐ Not confident
- ☐ Not very confident
- ☐ Neutral
- ☐ Somewhat confident
- ☐ Very confident

Section 3: Willingness to Adopt Generative AI

1. In your opinion, how important is it for employees in your company to receive formal training in usage of generative
- ☐ Not important at all
- ☐ Not really important
- ☐ Neutral
- ☐ Somewhat important
- ☐ Very important
2. How interested are you in learning about using generative AI in your current role?
- ☐ Not interested at all
- ☐ Not very interested
- ☐ Neutral
- ☐ Somewhat interested
- ☐ Very interested

Section 4: Generative AI Knowledge Interest

1. What AI topics would you be most interested in learning about?
- ☐ Text generation
- ☐ Image generation
- ☐ Speech generation

Section 5: Preferences on how to learn about Generative AI

1. What is your preferred way of learning about the possibilities of AI?
 - ☐ Attending webinars
 - ☐ Attending online courses
 - ☐ Reading articles
 - ☐ Watching videos
 - ☐ Hands-on experimentation (workshop)
 - ☐ Attending conferences or industry events
 - ☐ Listening to audio discussions/podcasts
 - ☐ Engaging in online AI communities
 - ☐ Following AI experts on social media
 - ☐ Not interested in learning about AI
 - ☐ Other (please specify):
2. How much time would you be willing to dedicate to learning about AI per week?
 - ☐ Less than 1 hour
 - ☐ 1 - 2 hours
 - ☐ 2 - 6 hours
 - ☐ More than 6 hours
 - ☐ One-time learning dedication (not weekly recurring)
3. What barriers, if any, do you experience in learning about AI for your role?
 - ☐ Lack of time to learn and practice
 - ☐ Insufficient access to learning resources
 - ☐ Complexity of AI concepts and tools
 - ☐ Lack of support or mentorship
 - ☐ High cost of training programs
 - ☐ Limited understanding of programming
 - ☐ Uncertainty about practical applications in my role
 - ☐ Rapidly changing landscape
 - ☐ Concern about ethical implications of AI
 - ☐ Unsure where to start
 - ☐ Lack of hands-on practice opportunities
 - ☐ No barriers
 - ☐ Other (please specify):